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# Sentiment Analysis – A Comparative Study


**Presentation By**

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# Inspiration and Objective

- **Inspiration:** To deepen our understanding of Natural Language Processing by exploring various sentiment analysis techniques. Driven by the need to compare traditional and modern approaches for practical sentiment classification.

- **Objective:** To implement and evaluate lexicon-based, machine learning, and deep learning models for sentiment analysis. To develop a comparative analysis of diverse models and feature extraction techniques on balanced datasets.

- **Key Goals:**
  - Implement and assess lexicon-based and classical ML sentiment classifiers.
  - Build and evaluate deep learning models including CNNs, LSTMs, and transformers.
  - Explore the effect of different feature engineering techniques like TF-IDF, Word2Vec, and GloVe.
  - Provide a comprehensive performance comparison to guide future sentiment analysis approaches.

# Presentation Outline

- Introduction
- Text Preprocessing
- Lexicon and Machine Learning
- Workflow
- Data
- Experiments
- Results and Conclusions
- Future Directions
- References

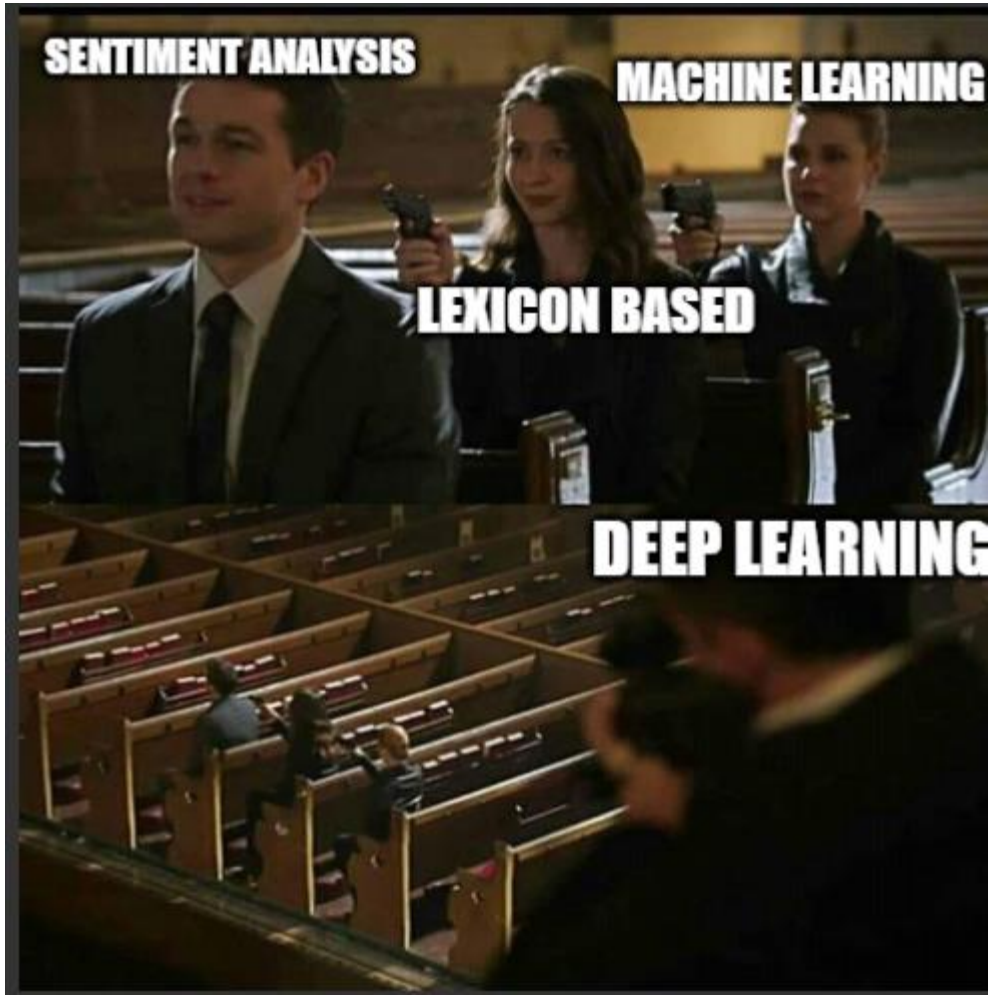
# Introduction

- **Explosion of Textual Data:** In today's digital age, massive amounts of text are generated daily from social media, reviews, blogs, and feedback systems, making manual analysis impractical.
- **Importance of Sentiment Analysis:** A key NLP technique, sentiment analysis helps automatically determine the emotional tone—positive, negative, or neutral—embedded in textual content.
- **Applications Across Domains:** Beyond education, sentiment analysis is used in business for customer reviews, in politics for public opinion mining, and in healthcare for analyzing patient feedback.
- **Evolution of Techniques:** The field has evolved from lexicon-based methods like VADER to machine learning models such as Naive Bayes and SVMs, and now to advanced deep learning and transformer-based models like BERT and GPT.

# Text Preprocessing

- **Level 1:** Tokenization, Stop words, Lemmatization
- **Level 2:** Bag of Words, TF IDF, Unigrams, Bigrams, N-Grams.
- **Level 3:** Word embeddings, word2vec, GloVe

# Lexicon and Machine Learning



# Lexicon and Machine Learning

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➢ **Lexicon Based**

- Do not require training on labeled data.
- Rely on predefined sentiment lexicons where each word is associated with a sentiment score.
- Sentiment of a sentence or document is calculated based on the aggregate scores of the words it contains.
- Simple, interpretable, and useful for domains where labeled data is scarce.

VADER =>

$$Compound\ Score = \frac{\sum_i s_i}{\sqrt{\sum_i s_i^2 + \alpha}}$$

# Lexicon and Machine Learning

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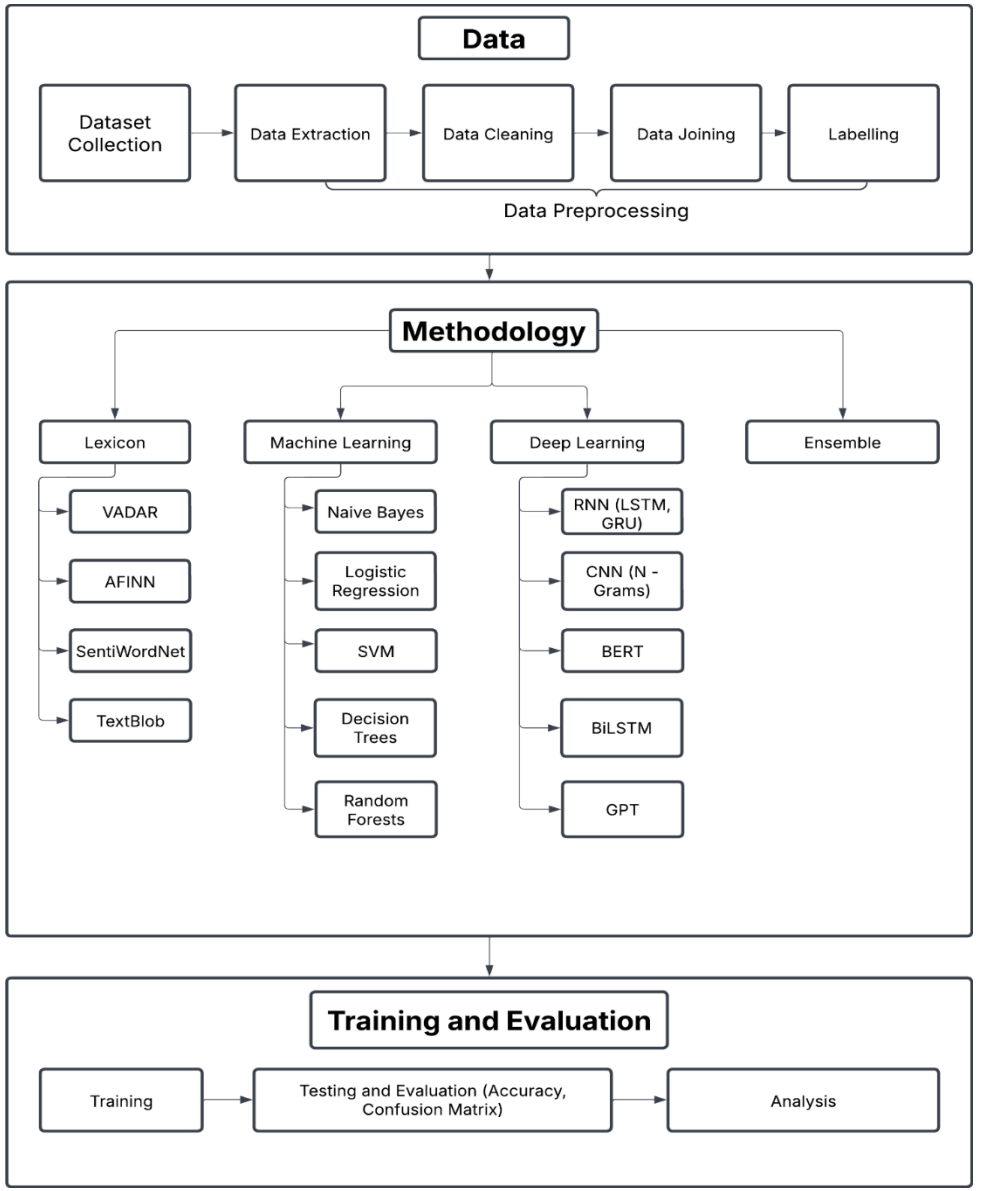
➢ **Machine learning Based**

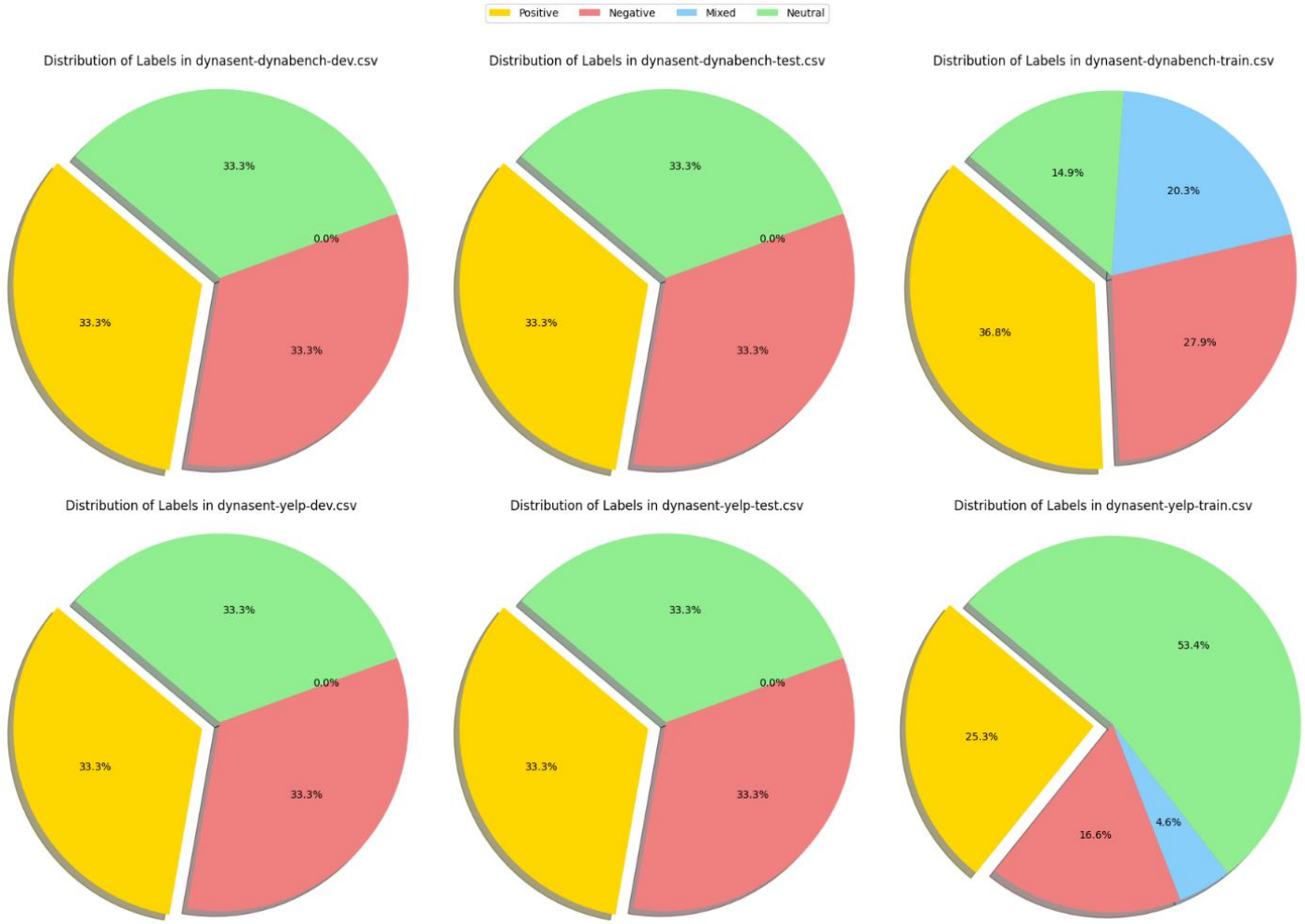
- Require labeled data for training models to learn sentiment patterns.
- Convert text into numerical features using techniques like CountVectorizer or TF-IDF.
- Learn patterns and decision boundaries using algorithms like:
  - Naive Bayes
  - Logistic Regression
  - Support Vector Machines (SVM)
  - Decision Tree / Random Forest

- Can generalize better than lexicon methods on large, diverse datasets.
- More flexible and customizable, but may require more computational resources.

# Workflow


Data
Dataset Collection
Data Extraction
Data Cleaning
Data Joining
Labelling
Data Preprocessing
Methodology
Lexicon
VADAR
AFINN
SentiWordNet
TextBlob
Machine Learning
Naive Bayes
Logistic Regression
SVM
Decision Trees
Random Forests
Deep Learning
RNN (LSTM, GRU)
CNN (N - Grams)
BERT
BiLSTM
GPT
Ensemble
Training and Evaluation
Training
Testing and Evaluation (Accuracy, Confusion Matrix)
Analysis

# Data


Positive
Negative
Mixed
Neutral
Distribution of Labels in dynasent-dynabench-dev.csv
33.3%
0.0%
33.3%
33.3%
Distribution of Labels in dynasent-dynabench-test.csv
33.3%
0.0%
33.3%
33.3%
Distribution of Labels in dynasent-dynabench-train.csv
14.9%
20.3%
36.8%
27.9%
Distribution of Labels in dynasent-yelp-dev.csv
33.3%
0.0%
33.3%
33.3%
Distribution of Labels in dynasent-yelp-test.csv
33.3%
0.0%
33.3%
33.3%
Distribution of Labels in dynasent-yelp-train.csv
53.4%
25.3%
4.6%
16.6%

# Data

| file_name | total_rows | positive_rows | negative_rows | mixed_rows | neutral_rows |
|---|---|---|---|---|---|
| dynasent-dynabench-dev.csv | 720 | 240 | 240 | 0 | 240 |
| dynasent-dynabench-test.csv | 720 | 240 | 240 | 0 | 240 |
| dynasent-dynabench-train.csv | 16399 | 6038 | 4579 | 3334 | 2448 |
| dynasent-yelp-dev.csv | 3600 | 1200 | 1200 | 0 | 1200 |
| dynasent-yelp-test.csv | 3600 | 1200 | 1200 | 0 | 1200 |
| dynasent-yelp-train.csv | 84388 | 21391 | 14021 | 3900 | 45076 |

| File_name | Total_rows | Positive | Negative | Neutral |
|---|---|---|---|---|
| Yelp_train | 42063 | 14021 | 14021 | 14021 |
| Yelp_test | 3600 | 1200 | 1200 | 1200 |
| Yelp_val | 3600 | 1200 | 1200 | 1200 |
| dynabench_train | 7344 | 2448 | 2448 | 2448 |
| dynabench_test | 720 | 240 | 240 | 240 |
| dynabench_val | 720 | 240 | 240 | 240 |
| combined_train | 49407 | 16469 | 16469 | 16469 |
| combined_test | 1440 | 480 | 480 | 480 |
| combined_val | 1440 | 480 | 480 | 480 |

# Experiments

- **Exp 1: Using VADER**

$$Compound\ Score = \frac{\sum_i s_i}{\sqrt{\sum_i s_i^2 + \alpha}}$$

- **Exp 2: Using AFINN**

$$Sentiment\ Score = \sum_{i=1}^{n} score(w_i)$$

- **Exp 3: Using SentiWordNet**

$$Sentiment\ Score\ (w) = \frac{1}{N} \sum_{i=1}^{N} (Pos_i(w) - Neg_i(w))$$

- **Exp 4: Using TextBlob**

# Experiments

- **Exp 5, 6: Naïve Bayes**

$$P(c,d) = \frac{P(c)\prod_{i=1}^{n} P(w_i|c)}{P(d)}$$

$$\hat{y} = argmax_c \left(\log P(c) + \sum_{i=1}^{n} log\ P(w_i|c)\right)$$

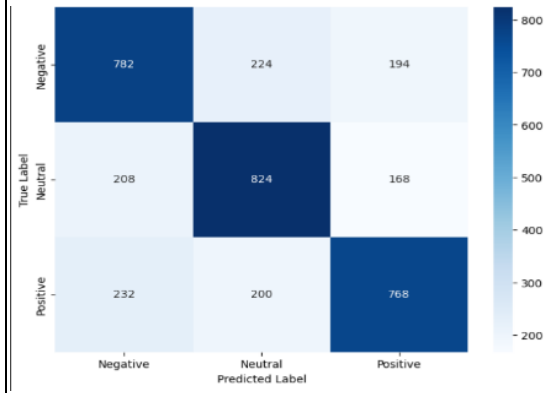
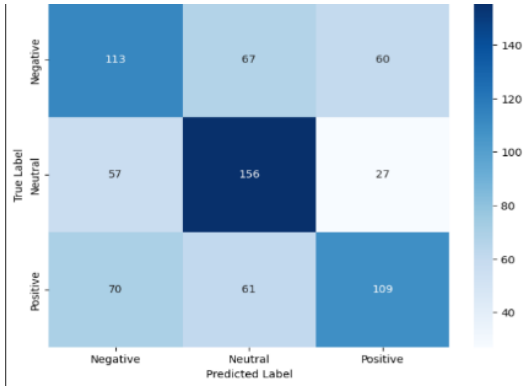
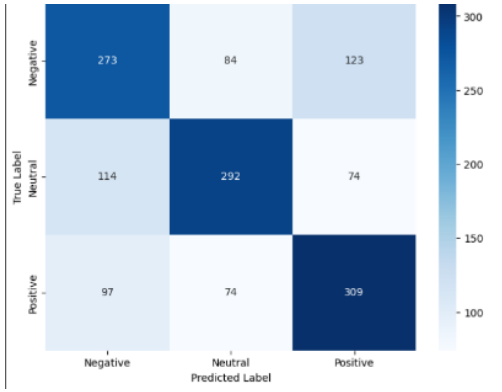
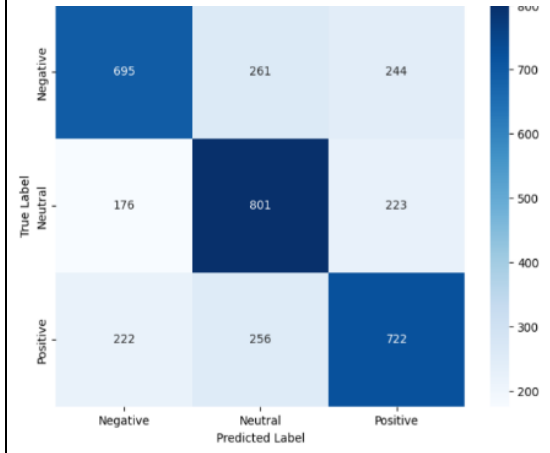
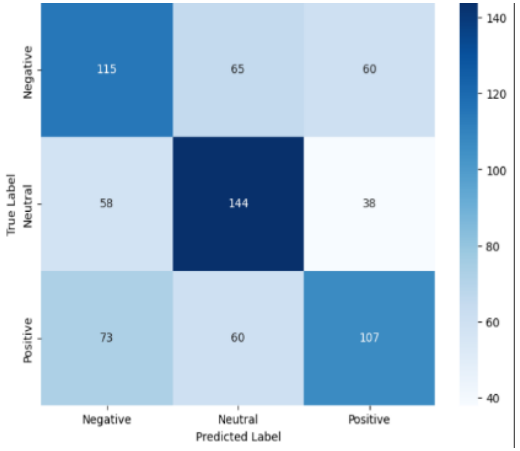
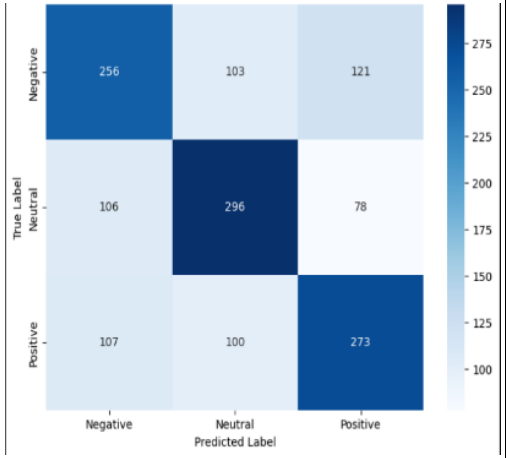
$$P(w_i|c) = \frac{N_{w_{i,c}} + 1}{\sum_j N_{w_{i,c}} + |V|}$$

- **Exp 7,8,9: Logistic Regression**
- **Exp 10: SVM**
- **Exp 11: Decision Tree**
- **Exp 12: Random Forest**

# Results

| Exp No. | Experiment Name | Yelp Dataset | Dynabench Dataset | Combined Dataset |
|---|---|---|---|---|
| 1 | Using VADER | 53 | 57 | 53 |
| 2 | Using AFINN | 52 | 57 | 53 |
| 3 | Using SentiWordNet | 43 | 44 | 43 |
| 4 | Using TextBlob | 50 | 54 | 51 |
| 5 | Naive Bayes from Scratch | 64 | 52 | 57 |
| 6 | Multinomial Naive Bayes | 65 | 53 | 57 |
| 7 | Logistic Regression 1 | 54 | 49 | 45 |
| 8 | Logistic Regression 2 | 60 | 50 | 56 |
| 9 | Logistic Regression 3 | 62 | 51 | 57 |
| 10 | SVM | 66 | 53 | 61 |
| 11 | Decision Tree | 55 | 49 | 52 |
| 12 | Random Forest | 62 | 54 | 59 |

# Results

| Exp Name | Yelp Dataset | Dynabench Dataset | Combined Dataset |
|---|---|---|---|
| SVM |  |  |  |
| Logistic Regression with TF-IDf features |  |  |  |

# Conclusion and Future Directions

- Among the machine learning models, Logistic Regression and SVM with TF-IDF feature extraction showed better performance in sentiment classification.
- Lexicon-based methods provided quick and interpretable results but lacked contextual understanding.
- Traditional machine learning models performed well with proper preprocessing and feature engineering.
- Explore deep learning approaches such as CNN, RNN, LSTM, and transformer-based models like BERT. These models can better capture semantic context, word dependencies, and long-range relationships in text. Fine-tuning pretrained models like BERT on domain-specific data may further improve accuracy.


Backspace
Thank You
Page Up
Home
Page Down
Insert
Shift